

SCIENTIFIC DISCOVERIES, NOBEL PRIZES AND SCIENTISTS

MINIMUM PRIZE EVIDENCE CARD – SCIENTIFIC CONTRIBUTION CLASSIFIER – NOBEL CATEGORY MAP – CITATION-TO-OUTCOME STAIRCASE – ATTRIBUTION AND CHRONOLOGY RAIL – PHYSICS 2025 QUANTUM-CIRCUIT MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

MINIMUM PRIZE EVIDENCE CARD

MINIMUM PRIZE EVIDENCE CARDCATEGORY + YEAROFFICIAL LAUREATE SPELLINGOFFICIAL CITATIONMECHANISM + AWARD-TIME AFFILIATIONINDIA LINKSTATUS DATEONLY IF SOURCED

CONCEPT / ARCHITECTURE

- CATEGORY + YEAR
- OFFICIAL LAUREATE SPELLING

MECHANISM / APPLICATION

- OFFICIAL CITATION / MOTIVATION
- MECHANISM + AWARD-TIME AFFILIATION

READINESS / STATUS LIMIT

- BIRTH / INDIA LINK / STATUS DATE → only if sourced
- CATEGORY + YEAR

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BIRTH / INDIA LINK / STATUS DATE → only if sourced.

01

STAGE 01

SCIENTIFIC CONTRIBUTION CLASSIFIER

SCIENTIFIC CONTRIBUTION CLASSIFIEREXPLANATION AND PREDICTIONRECORDED PHENOMENONCONTROLLED TESTESTABLISHED PHENOMENON OR RELATIONDEVICE, PROCESS OR METHOD

CONCEPT / ARCHITECTURE

- THEORY → explanation and prediction
- OBSERVATION → recorded phenomenon

MECHANISM / APPLICATION

- EXPERIMENT → controlled test
- DISCOVERY → established phenomenon or relation

READINESS / STATUS LIMIT

- INVENTION / DEVELOPMENT → device, process or method
- THEORY → explanation and prediction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INVENTION / DEVELOPMENT → device, process or method.

02

STAGE 02

NOBEL CATEGORY MAP

NOBEL CATEGORY MAPPHYSIOLOGY OR MEDICINEORIGINAL FRAMEWORKECONOMIC SCIENCESLATER MEMORIAL PRIZEAWARDING INSTITUTIONPERSON OR ORGANISATION POSSIBLE

CONCEPT / ARCHITECTURE

- PHYSIOLOGY OR MEDICINE
- PEACE → original framework

MECHANISM / APPLICATION

- ECONOMIC SCIENCES → later memorial prize
- AWARDING INSTITUTION → category-specific

READINESS / STATUS LIMIT

- PEACE → person or organisation possible
- PHYSIOLOGY OR MEDICINE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AWARDING INSTITUTION → category-specific; PEACE → person or organisation possible.

03

STAGE 03

CITATION-TO-OUTCOME STAIRCASE

CITATION-TO-OUTCOME STAIRCASEOFFICIAL CITATIONPOSSIBLE APPLICATIONENGINEERING + COMMERCIALIZATIONMEASURED POLICYSOCIAL OUTCOME

CONCEPT / ARCHITECTURE

- OFFICIAL CITATION
- DISCOVERY / MECHANISM

MECHANISM / APPLICATION

- POSSIBLE APPLICATION
- ENGINEERING + COMMERCIALIZATION

READINESS / STATUS LIMIT

- MEASURED POLICY / SOCIAL OUTCOME
- OFFICIAL CITATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENGINEERING + COMMERCIALIZATION; MEASURED POLICY / SOCIAL OUTCOME.

04

STAGE 04

ATTRIBUTION AND CHRONOLOGY RAIL

ATTRIBUTION AND CHRONOLOGY RAILPRIOR THEORYLATER AND SEPARATEAWARD YEARDISCOVERY YEAR AUTOMATICALLY

CONCEPT / ARCHITECTURE

- PRIOR THEORY → INSTRUMENT → EXPERIMENT
- PUBLICATION → REPLICATION → AWARD

MECHANISM / APPLICATION

- PATENT / PRODUCT / DEPLOYMENT → later and separate
- LAUREATE != institution

READINESS / STATUS LIMIT

- AWARD YEAR != discovery year automatically
- PRIOR THEORY → INSTRUMENT → EXPERIMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PATENT / PRODUCT / DEPLOYMENT → later and separate.

05

STAGE 05

PHYSICS 2025 QUANTUM-CIRCUIT MAP

PHYSICS 2025 QUANTUM-CIRCUIT MAPSUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTIONCOLLECTIVE MACROSCOPIC STATEQUANTUM TUNNELLINGSPECIFIC ENERGY AMOUNTSFAULT-TOLERANT QUANTUM COMPUTER

CONCEPT / ARCHITECTURE

- SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION
- COLLECTIVE MACROSCOPIC STATE

MECHANISM / APPLICATION

- BARRIER → quantum tunnelling
- SPECIFIC ENERGY AMOUNTS → quantisation

READINESS / STATUS LIMIT

- DEMONSTRATION != fault-tolerant quantum computer
- SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPECIFIC ENERGY AMOUNTS → quantisation; DEMONSTRATION != fault-tolerant quantum computer.

06

STAGE 06

JOHN CLARKE CLUE RESOLVER

JOHN CLARKE CLUE RESOLVERCAMBRIDGE, UNITED KINGDOMUNIVERSITY OF CALIFORNIA, BERKELEYPHYSICS 2025ONE-THIRD WITH DEVORET AND MARTINISMATCH ALL CLUESTHEN IDENTIFY

CONCEPT / ARCHITECTURE

- BORN → Cambridge, United Kingdom
- AFFILIATION → University of California, Berkeley

MECHANISM / APPLICATION

- CATEGORY / YEAR → Physics 2025
- SHARE → one-third with Devoret and Martinis

READINESS / STATUS LIMIT

- MATCH ALL CLUES → then identify
- BORN → Cambridge, United Kingdom

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SHARE → one-third with Devoret and Martinis; MATCH ALL CLUES → then identify.

07

STAGE 07

CHEMISTRY 2025 MOF ARCHITECTURE

CHEMISTRY 2025 MOF ARCHITECTUREMETAL IONSORGANIC MOLECULESPOROUS CRYSTALBUILDING-BLOCK CHANGETUNABLE FUNCTION

CONCEPT / ARCHITECTURE

- METAL IONS → nodes
- ORGANIC MOLECULES → linkers

MECHANISM / APPLICATION

- POROUS CRYSTAL → cavities
- BUILDING-BLOCK CHANGE → tunable function

READINESS / STATUS LIMIT

- CAPABILITY / APPLICATION != commercialization
- METAL IONS → nodes

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BUILDING-BLOCK CHANGE → tunable function; CAPABILITY / APPLICATION != commercialization.

08

STAGE 08

MEDICINE 2025 TOLERANCE PATHWAY

MEDICINE 2025 TOLERANCE PATHWAYCENTRAL TOLERANCETHYMIC CONTROL LAYERREGULATORY T CELLSPERIPHERAL CONTROLREGULATORY-T-CELL DEVELOPMENTSELF-TISSUE PROTECTIONCLINICAL TRIAL

STARTING CONDITION / INPUT

- CENTRAL TOLERANCE → thymic control layer
- REGULATORY T CELLS → peripheral control

SCIENTIFIC OR ENGINEERING MECHANISM

- Foxp3 → regulatory-T-cell development
- SELF-TISSUE PROTECTION → mechanism

OUTPUT / FAILURE BOUNDARY

- CLINICAL TRIAL != approved therapy
- CENTRAL TOLERANCE → thymic control layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SELF-TISSUE PROTECTION → mechanism; CLINICAL TRIAL != approved therapy.

09

STAGE 09

MICRORNA REGULATION RAIL

MICRORNA REGULATION RAILDNA --TRANSCRIPTION-- mRNA microRNACOMPLEMENTARY mRNA BINDINGPROTEIN PRODUCTIONINHIBITED MRNA DEGRADATIONPOSSIBLE ROUTEDIAGNOSTIC OR MEDICINE

CONCEPT / ARCHITECTURE

- DNA --transcription-- mRNA microRNA → complementary mRNA binding
- PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route

MECHANISM / APPLICATION

- DISCOVERY != diagnostic or medicine
- DNA --transcription-- mRNA microRNA → complementary mRNA binding

READINESS / STATUS LIMIT

- PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route
- DISCOVERY != diagnostic or medicine

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route.

10

STAGE 10

INDIA-LINKED SCIENTIST CARDS

INDIA-LINKED SCIENTIST CARDSRAMAN 1930LIGHT SCATTERINGRAMAN EFFECTKHORANA 1968GENETIC CODEPROTEIN SYNTHESISRAMAKRISHNAN 2009

CONCEPT / ARCHITECTURE

- RAMAN 1930 → light scattering / Raman effect
- KHORANA 1968 → genetic code / protein synthesis

MECHANISM / APPLICATION

- RAMAKRISHNAN 2009 → ribosome structure and function
- BIRTHPLACE != award-time affiliation

READINESS / STATUS LIMIT

- CITATION != every later application
- RAMAN 1930 → light scattering / Raman effect

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: KHORANA 1968 → genetic code / protein synthesis.

11

STAGE 11

PYQ AND FUTURE-STATUS FIREWALL

PYQ AND FUTURE-STATUS FIREWALL2026 Q81DIRECT BIOGRAPHICAL PRIZE ROUTEBLUE LEDTOPIC 21 DIRECT SCIENCE OWNERTOPIC 26ATTRIBUTION + CITATION + MATURITY SUPPORT2025 OFFICIAL AWARDS

SYSTEM / DEVICE / INSTITUTION

- 2026 Q81 → direct biographical prize route
- BOSE-EINSTEIN / BLUE LED → Topic 21 direct science owner

DECISIVE TECHNICAL DISTINCTION

- TOPIC 26 → attribution + citation + maturity support
- 2025 OFFICIAL AWARDS → latest completed science anchors

STATUS / UNIT / EVIDENCE LIMIT

- 4 SEP 2026 → no invented 2026 laureates
- 2026 Q81 → direct biographical prize route

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 4 SEP 2026 → no invented 2026 laureates.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENTUSE LAUREATE EXAMPLES TO ILLUSTRATE LONG-HORIZON PUBLIC RESEARCH, INSTRUMENTATIONMOBILITY AND THE PATH FROM BASIC SCIENCE TO STRATEGICAVOID "NOBEL COUNT" NATIONALISM.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: Use laureate examples to illustrate long-horizon public research, instrumentation, international
- mobility and the path from basic science to strategic technology.
- CAUTION: Avoid "Nobel count" nationalism.
- A stronger answer examines research funding, universities, laboratory autonomy, interdisciplinary teams and translation capacity.

READINESS, UNIT OR STATUS LIMIT

- CAUTION: Connect Topic 24's funding architecture with the specialist science file that explains the discovery.
- Subject: Science & Technology | Tier: Advanced | GS Paper: GS-III + Prelims.
- Purpose: Convert prize current affairs into deeper scientific reasoning without making optional enrichment necessary for objective-question coverage.
- Analytical lens CAUTION: Major prizes reveal how science moves from a counter-intuitive

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- mechanism to an experimental demonstration and then to a technology platform.
- Deeper distinctions | Distinction | Why it matters | |---| | Discovery vs invention | A discovery establishes a phenomenon
- an invention creates a device or process.
- Prize citations may recognise either kind of contribution.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.